

1.

537 The function f is defined by $f(x) = 8(3)^{-\frac{x}{7}}$. What is the value of $f(14)$?

A. -72

B. -48

C. $\frac{1}{576}$

D. $\frac{8}{9}$

2.

538 In 2010, a skating team had a total of 51 skaters, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced skaters on the team increased by approximately 59%, and the number of intermediate skaters on the team increased by approximately 48%. The total number of skaters on the team increased by approximately 53%. Which equation best represents this situation, where a represents the number of advanced skaters on the team in 2010 and b represents the number of intermediate skaters on the team in 2010?

A. $1.48a + 1.59b = 51(1.53)$

B. $1.59a + 1.48b = 51(1.53)$

C. $1.53a + 0.59b = 51(1.48)$

D. $1.59a + 1.53b = 51(1.48)$

3.

540 An artist made 10 pieces of pottery using 36 pounds of clay. The pottery included only bowls and vases. Each bowl used 4 pounds of clay and each vase used 2 pounds of clay. How many vases did the artist make?

A. 2

B. 4

C. 6

D. 8

4.

548

$$\begin{aligned}y - x &= 47 \\ y &= x^2 - 45x\end{aligned}$$

The graphs of the equations in the given system intersect at the point (x, y) in the xy -plane. What is a possible value of y ?

A. 47

B. 48

C. 93

D. 94

4.

549

Line p is defined by $kx + 7y = 16$. Line p passes through the points $(r, 0)$ and $(0, t)$ in the xy -plane, where k , r , and t are nonzero constants and r is not equal to t . Which expression represents the value of k ?

A. $\frac{16}{t}$

B. $\frac{16}{r}$

C. $\frac{7}{t}$

D. $\frac{7}{r}$

5.

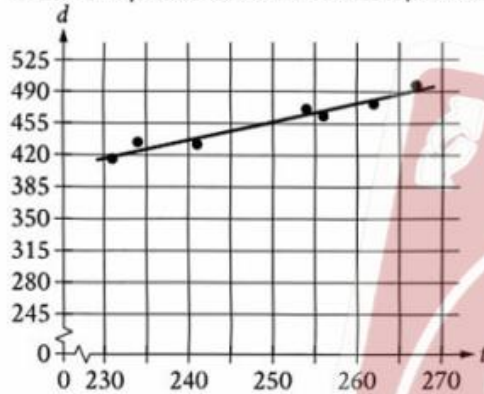
550

$$nx + 3t = -3x + 4n$$

In the given equation, n and t are constants. The equation has no solution. What **CANNOT** be the value of $n + t$?

6.

55 The scatterplot shows the relationship between two variables, t and d .



Which of the following equations is the most appropriate linear model for the data shown?

A. $d = -48.1 + 2.02t$

B. $d = 148.1 + 2.02t$

C. $d = 371.8 + 2.02t$

D. $d = 406.8 + 2.02t$

7.

552

	Volume (cubic units)
Right circular cylinder A	32π
Right circular cylinder B	864π

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 2 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

8.

560

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y , in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 2.75 inches, it has an estimated breaking strength of 9,528.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 8.50 inches?

9.

563

$$(x + k)^2 + 31 = 31$$

In the given equation, k is a constant. How many distinct real solutions does this equation have?

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

10.

567

During the first 8.00 seconds after a car started moving, its speed increased to 11.0 meters per second. From 8.00 seconds to 14.0 seconds after the car started moving, its speed increased from 11.0 meters per second to 23.0 meters per second. To the nearest hundredth, what is the positive difference between the average rate of change of the speed of the car, in meters per second per second, during the first 8.00 seconds after the car started moving and the average rate of change of the speed of the car, in meters per second per second, from 8.00 seconds to 14.0 seconds after the car started moving?

11

568

$$f(x) = 4x^2 + 56x + 197$$

The function g is defined by $g(x) = f(x + 6)$. What is the minimum value of $g(x)$?

- A. -13
- B. -7
- C. 1
- D. 7

12

624

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(9x - 650)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)^\circ$. What is the value of w ?

13.

62

$$y = 4x^2 - bx - 5$$

Which of the following equations is equivalent to the given equation, where b is a positive constant?

A. $y = 4\left(x - \frac{b}{8}\right)^2 - 5 - \frac{b^2}{16}$

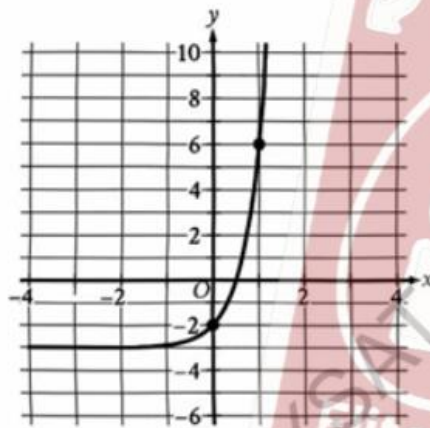
B. $y = 4\left(x + \frac{b}{8}\right)^2 - 5$

C. $y = 4\left(x - \frac{b}{8}\right)^2 - 5$

D. $y = 4\left(x + \frac{b}{8}\right)^2 - 5 - \frac{b^2}{16}$

14.

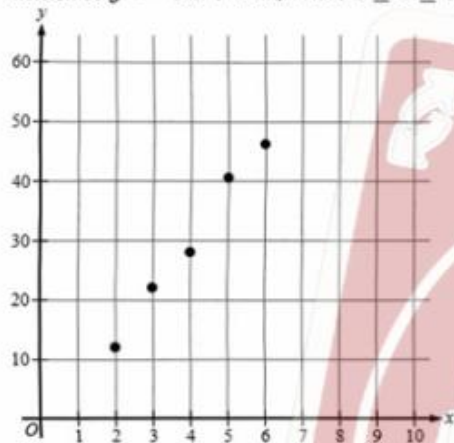
62



The equation of the graph shown is $y = a^x + b$, where a and b are constants. What is the value of $a - b$?

15.

- 618 The scatterplot shows data set A, which consists of the weights y , in pounds, of a Labrador retriever puppy at various ages, x , in months. The equation of a line of best fit for the relationship in data set A can be written as $y = -5.0 + 8.7x$, where $2 \leq x \leq 6$.



The puppy was weighed again at 9 months old and weighed 52 pounds. Data set B consists of all the data points in data set A as well as the data point (9, 52). The equation of a line of best fit for data set B can be written as $y = r + sx$, where r and s are constants and $2 \leq x \leq 9$. Assuming the equations of the lines of best fit are calculated in the same way, which of the following is the best estimate for the value of s ?

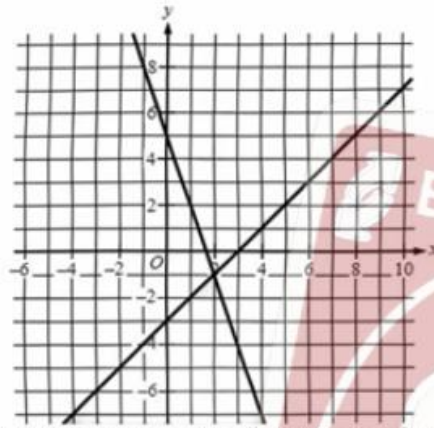
- A. 5.8
- B. 8.7
- C. 13.7
- D. 17.7

16

- 617 On a plot of land, 48.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 28.5% of the square footage of the farmland, and there are buildings on exactly 11.0% of the square footage of the pasture. If there are buildings on exactly $p\%$ of the square footage of the plot of land, what is the value of p ?

17

614



The graphs of $y = -3x + 5$ and $y = x - 3$ are shown. Point P (not shown) has coordinates $(2, -3)$ and lies in the shaded region. The coordinates of P satisfy which of the following inequalities?

- I. $y < -3x + 5$
- II. $y > x - 3$

A. I only

B. II only

C. I and II

D. Neither I nor II

18.

605

x	$g(x)$
1	34
2	27
3	20
4	13

For the linear function g , the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b ?

A. 7

B. 17

C. 34

D. 41

19.

- 603 In 2010, a skating team had a total of 34 skaters, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced skaters on the team increased by approximately 67%, and the number of intermediate skaters on the team increased by approximately 44%. The total number of skaters on the team increased by approximately 56%. Which equation best represents this situation, where a represents the number of advanced skaters on the team in 2010 and b represents the number of intermediate skaters on the team in 2010?

A. $1.44a + 1.67b = 34(1.56)$

B. $1.67a + 1.44b = 34(1.56)$

C. $1.56a + 0.67b = 34(1.44)$

D. $1.67a + 1.56b = 34(1.44)$

20.

- 602 In the xy -plane, lines k and j are perpendicular and intersect at the point $(5, 26)$. Line j passes through the origin, and line k passes through the point $(p, 0)$, where p is a constant. What is the value of p ?

A. -130.2

B. -129.8

C. 139.8

D. 140.2

21.

- 600 A store sells bottles of water for \$0.88 each and makes a profit of \$0.42 on each bottle sold. During the first three days of a week, the store sold 108 bottles of water. The rest of the week, the store sold w bottles of water. The store made a total profit of \$135.66 from selling bottles of water during the week. Which equation represents this situation?

A. $(0.88 - 0.42)(w + 108) = 135.66$

B. $(0.88 - 0.42)(w - 108) = 135.66$

C. $0.42(w + 108) = 135.66$

D. $0.42(w - 108) = 135.66$

22.

- 597 Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is 58 square centimeters (cm^2), and the surface area of right rectangular prism Y is $1,450 \text{ cm}^2$. The volume of right rectangular prism Y is $1,500 \text{ cubic centimeters (cm}^3\text{)}$. What is the sum of the volumes, in cm^3 , of right rectangular prism X and right rectangular prism Y?

23.

- 598 The table summarizes the distribution of age and assigned group for 120 participants in a study.

	0–9 years	10–19 years	20+ years	Total
Group A	15	16	9	40
Group B	5	9	26	40
Group C	20	15	5	40
Total	40	40	40	120

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

A. $\frac{5}{24}$

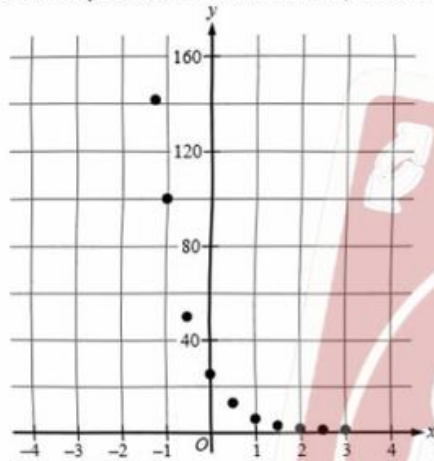
B. $\frac{5}{16}$

C. $\frac{3}{5}$

D. $\frac{5}{9}$

24.

593 The scatterplot shows the relationship between x and y for the 10 data points in data set G.



Data set H is created by multiplying the y -value of each data point from data set G by 20. Which of the following equations is the most appropriate model for data set H?

A. $y = 20(0.30)^x + 20$

B. $y = 26(0.30)^x$

C. $y = 26(0.30)^x + 20$

D. $y = 522(0.30)^x$

25.

590 A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was 6,950% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 32% of the number of undergraduate students surveyed. If there were 4,448 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

26.

584 The given function f estimates the distance a train has traveled, in kilometers, from a station in a certain city x hours after crossing the city border. What is the best interpretation of 8 in this context?

A. Between the station and the city border, the train traveled an estimated total distance of 8 kilometers.

B. After crossing the city border, the train traveled at an estimated speed of 8 kilometers per hour.

C. After crossing the city border, the train traveled an estimated total distance of 8 kilometers.

D. Between the station and the city border, the train traveled at an estimated speed of 8 kilometers per hour.

27

- 356 Nina is selling coins from her coin collection. When she started selling the coins, there were 2,600 coins in the collection. At the end of each week for 3 weeks after Nina started selling the coins, the number of coins in the collection had decreased by $\frac{1}{2}$ of the coins in the collection at the end of the previous week. Which equation gives the number of coins, c , in the collection w weeks after Nina started selling the coins, where $0 \leq w \leq 3$?

A. $c = w^{\frac{1}{2}}$

B. $c = 2,600^w$

C. $c = 1,300\left(\frac{1}{2}\right)^w$

D. $c = 2,600\left(\frac{1}{2}\right)^w$

28.

- 364 For the polynomial function f , the graph of $y = f(x)$ in the xy -plane passes through the points $(-5, 0)$, $(4, 0)$, and $(9, 0)$. Which of the following must be a factor of $f(x)$?

A. $x + 4$

B. $x + 9$

C. $x - 4$

D. $x - 5$

29.

- 368 Circles A and B are graphed in the xy -plane. Circle A is represented by the equation $(x - 9)^2 + (y + 9)^2 = 81$ and intersects the x -axis at the point (r, s) . Circle B has its center at (r, s) and has a radius of the same length as circle A . Which equation represents circle B ?

A. $x^2 + (y - 9)^2 = 81$

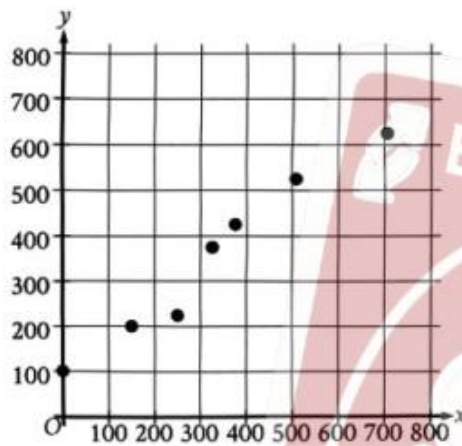
B. $x^2 + (y + 9)^2 = 81$

C. $(x - 9)^2 + y^2 = 81$

D. $(x + 9)^2 + y^2 = 81$

30.

369 The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

A. $y = -0.80x$

B. $y = 89.87x$

C. $y = 0.80 + 89.87x$

D. $y = 89.87 + 0.80x$

31

371

$$RS = 792$$

$$ST = 160$$

$$TR = 808$$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

A. $\frac{20}{101}$

B. $\frac{20}{99}$

C. $\frac{99}{101}$

D. $\frac{99}{20}$

32.

372

$$f(x) = 26(2)^x + 4(2)^x$$

The function f is defined by the given equation. Which of the following equivalent forms of f displays, as a coefficient or a base, the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

A. $f(x) = 104(2)^{x-2} + 16(2)^{x-2}$

B. $f(x) = 2(13(2)^x + 2(2)^x)$

C. $f(x) = 26(4)^{\frac{x}{2}} + 4(4)^{\frac{x}{2}}$

D. $f(x) = 30(2)^x$

33.

373

In a set of three consecutive integers, where the integers are ordered from least to greatest, the first integer is represented by x . The sum of 4 and the second integer is less than the product of 17 and the third integer. Which inequality represents this situation?

A. $4 + (x + 1) > 17(x + 2)$

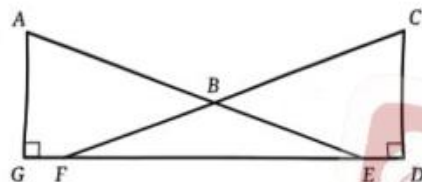
B. $4 + (x + 1) < 17(x + 2)$

C. $4 + (x + 2) > 17(x + 3)$

D. $4 + (x + 2) < 17(x + 3)$

34.

403



Note: Figure not drawn to scale.

In the figure, triangle AEG is congruent to triangle CFD , where E corresponds to F . The measure of angle ABF is 46° . What is the measure, in degrees, of angle DCF ?

A. 23

B. 44

C. 46

D. 67

35.

- 404 An ice cube is floating in water. The portion of the ice cube below the water's surface has a volume of 12.28 cubic centimeters and makes up 91.67% of the total volume of the ice cube. What is the volume, in cubic centimeters, of the portion of the ice cube above the water's surface?

36.

- 421 The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 8)$ and passes through the points $(2, 47)$ and $(-1, 164)$. What is the value of $f(-2) - f(0)$?

- A. 125
- B. 203
- C. 312
- D. 359

37.

- 610 Micah deposited a total of \$3,600 into two retirement accounts last year by depositing x dollars once each month into one account and y dollars twice each month into the other account. Which equation represents this situation?

- A. $3,600 = 12x + 12y$
- B. $3,600 = 12x + 24y$
- C. $3,600 = 24x + 12y$
- D. $3,600 = 24x + 24y$

38.

- 620 A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$120.00, zero products would be sold. For each \$1.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$1,966.50, which of the following could be the number of products sold? (revenue = price $\times$ number of products sold)

- A. 23
- B. 40
- C. 1,231
- D. 2,400

39.

- 625 The equation $d = 6t + 270$ gives the distance d , in feet, from a reference point on the surface of Mars to a Mars rover, where t is the number of minutes after the start of a period of observation. By how many feet does the Mars rover's distance from the reference point increase over a period of 20 minutes?

A. 390

B. 290

C. 120

D. 6

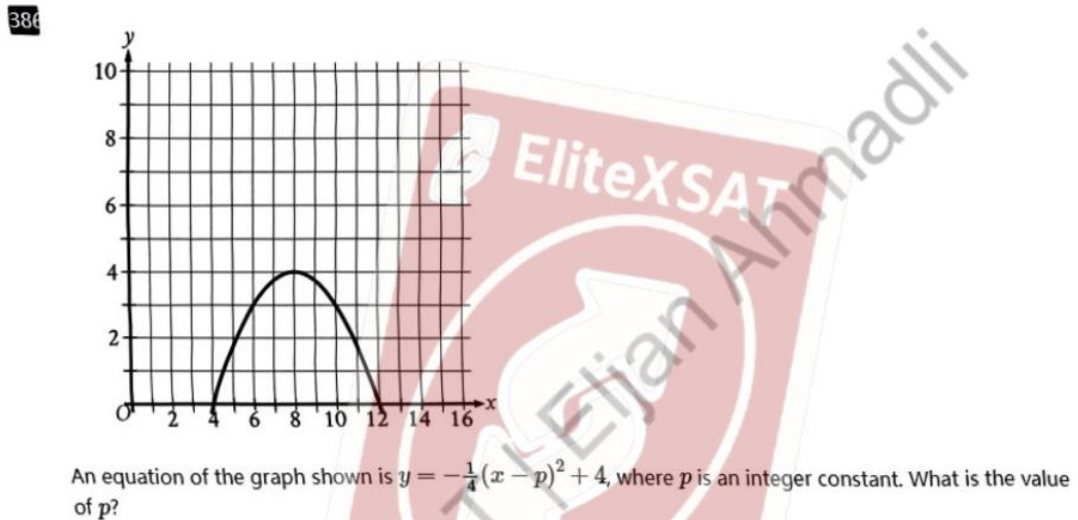
40.

- 625 In the given equation, b is an integer. The equation has no real solutions. What is the least possible value of b ?
- $$-x^2 + bx - 49 = 0$$

41.

- 387 A model predicts that a certain elephant calf weighed 249 pounds when it was born and that the calf gained 2 pounds per day in its first year of life. This model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the elephant calf x days after it was born, and a and b are constants. What is the value of a ?

42.



43.

- 388 The point $(0, 0)$ lies on a circle in the xy -plane. An equation of this circle is $x^2 + (y + 6)^2 = p + 3$, where p is a positive constant. What is the value of p ?

44.

399

Data Set A

Value Frequency

c 12

$2c$ 21

$3c$ 30

Data Set B

Value Frequency

c 30

$2c$ 21

$3c$ 12

The frequency tables represent data sets A and B, where c is a negative integer constant. The mean of data set A is r and the mean of data set B is q . What is the value of $\frac{r}{q}$?

A. $-\frac{4}{3}$

B. 1

C. $\frac{4}{3}$

D. There is not enough information to determine the value of $\frac{r}{q}$.